

Rodrigue's formula :-

Legendre's Eqn :-

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0, \quad n \in \mathbb{Z}_{\geq 0}$$

- $x=0$ is an ordinary point

$$y = a_0 y_1(x) + a_1 y_2(x)$$

$\downarrow$
contains only even powers of x

$\rightarrow$ contains only odd powers of x

If n is even, $y_1(x)$ is a polynomial of degree n
If n is odd, $y_2(x)$ is a polynomial of degree n
 $\rightarrow$ Legendre's polynomials

$$P_n(x) \xrightarrow{n \text{ is even}} a_0 + a_2 x^2 + a_4 x^4 + \dots + a_n x^n$$

$$P_n(x) \xrightarrow{n \text{ is odd}} a_1 x + a_3 x^3 + \dots + a_n x^n$$

Rodrigue's formula :-

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2-1)^n$$

$$P_0(x) = \frac{1}{2^0 \cdot 0!} (x^2-1)^0 = 1, \quad P_1(x) = \frac{1}{2^1 \cdot 1!} \frac{d}{dx} (x^2-1) = \frac{2x}{2} = x$$

$$P_2(x) = \frac{1}{2^2 \cdot 2!} \frac{d^2}{dx^2} (x^2-1)^2 = \frac{1}{2^2 \cdot 2!} \frac{d}{dx} (2(x^2-1) \cdot 2x) = \frac{1}{2^2 \cdot 2!} \frac{d}{dx} (4x^3 - 4x)$$

$$P_2(x) = \frac{1}{2^2 \cdot 2!} (12x^2 - 4) = \frac{3}{2} x^2 - \frac{1}{2}$$

Example ① $2x^2y'' + (x^2-x)y' + y = 0$, find power series solution around $x=0$.

$$P(x) = \frac{x^2-x}{2x^2}$$

$$Q(x) = \frac{1}{2x^2}$$

$$\lim_{x \rightarrow 0} P(x) = \text{Not defined.}$$

$$\lim_{x \rightarrow 0} Q(x) \rightarrow \infty$$

$\Rightarrow x=0$ is not an ordinary point.

$$\lim_{x \rightarrow 0} x \cdot \frac{(x^2-x)}{2x^2} = \lim_{x \rightarrow 0} \frac{(x-1)}{2} = -\frac{1}{2} = p_0$$

$$\lim_{x \rightarrow 0} x^2 \cdot \frac{1}{2x^2} = \frac{1}{2} = q_0$$

Indicial Eqn

$$r(r-1) + p_0r + q_0 = 0$$

Indicial Eqn :- $r^2 - r - \frac{1}{2}r + \frac{1}{2} = 0 \Rightarrow 2r^2 - 3r + 1 = 0$
 $r = \frac{1}{2}, 1$

Let $y = x^r \sum_{m=0}^{\infty} a_m x^m$ be a solution of ①

$$y' = \sum_{m=0}^{\infty} (m+r) a_m x^{m+r-1}$$

$$y'' = \sum_{m=0}^{\infty} (m+r)(m+r-1) a_m x^{m+r-2}$$

$$2x^2y'' + (x^2-x)y' + y = 0$$

$$2 \sum_{m=0}^{\infty} (m+r)(m+r-1) a_m x^{m+r} + \sum_{m=0}^{\infty} (m+r) a_m x^{m+r+1} + \sum_{m=0}^{\infty} a_m x^{m+r} = 0$$

$$2 \sum_{m=0}^{\infty} (m+k)(m+k-1) a_m x^{m+k}$$

$$- \sum_{m=0}^{\infty} (m+k) a_m x^{m+k}$$

$$+ \sum_{m=0}^{\infty} a_m x^{m+k} = 0$$

$$\Rightarrow 2(k)(k-1) a_0 x^k + 2 \sum_{m=1}^{\infty} (m+k)(m+k-1) a_m x^{m+k} + \sum_{m=0}^{\infty} (m+k) a_m x^{m+k} - k a_0 x^k - \sum_{m=1}^{\infty} (m+k) a_m x^{m+k} + a_0 x^k + \sum_{m=1}^{\infty} a_m x^{m+k} = 0$$

$$\Rightarrow (2k^2 - 2k - k + 1) a_0 x^k + \sum_{m=1}^{\infty} [2(m+k)(m+k-1) a_m - (m+k) a_m + a_m + (m+k-1) a_{m-1}] x^{m+k} = 0$$

$$* \sum_{m=0}^{\infty} (m+k) a_m x^{m+k+1} = \sum_{m=1}^{\infty} (m+k-1) a_{m-1} x^{m+k}$$

$$* 2k^2 - 3k + 1 = 0 \Rightarrow k = \frac{1}{2}, 1$$

$$\bullet [2(m+k)(m+k-1) - (m+k) + 1] a_m = -(m+k-1) a_{m-1}$$

$$a_m = \frac{-(m+k-1)}{2(m+k)(m+k-1) - (m+k) + 1} a_{m-1}$$

$$k = \frac{1}{2}$$

$$a_m = \frac{-(m - \frac{1}{2})}{2(m + \frac{1}{2})(m - \frac{1}{2}) - (m + \frac{1}{2}) + 1} a_{m-1}$$

$$\left\{ \begin{aligned} & 2(m + \frac{1}{2})(m - \frac{1}{2}) - m - \frac{1}{2} + 1 = -(m - \frac{1}{2}) \\ & (m + \frac{1}{2}) [2m + 1 - 1] \end{aligned} \right.$$

$$a_m = \frac{-a_{m-1}}{2m}$$

$$\underline{m=1} \quad a_1 = -\frac{a_0}{2}, \quad a_2 = -\frac{a_1}{4} = \frac{a_0}{8}, \quad a_3 = -\frac{a_2}{6} = -\frac{a_0}{48}$$

$$y = x^{1/2} \cdot a_0 \left[1 + \left(-\frac{1}{2}\right)x + \left(\frac{1}{8}\right)x^2 - \frac{1}{48}x^3 + \dots \right]$$

$$a_m = \frac{-(m+k-1)}{2(m+k)(m+k-1) - (m+k) + 1} a_{m-1}$$

$$\boxed{k=1}$$

$$a_m = \frac{-m a_{m-1}}{2(m+1)m - m - 1 + 1} = \frac{-m}{2m^2 + 2m - m} a_{m-1}$$

$$a_m = \frac{-m}{m(2m+1)} a_{m-1}$$

$$a_m = \frac{-1}{2m+1} a_{m-1}$$

$$\underline{m=1} \quad a_1 = -\frac{1}{3}a_0, \quad a_2 = -\frac{1}{5}a_1 = \frac{1}{15}a_0, \quad a_3 = -\frac{a_2}{7} = -\frac{a_0}{7 \cdot 15}$$

$$y_2(x) = a_0 x^1 \left(1 - \frac{1}{3}x + \frac{1}{3 \cdot 5}x^2 - \frac{1}{3 \cdot 5 \cdot 7}x^3 + \dots \right)$$

Complete solution

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

Example

$$x(x-1)y'' + (3x-1)y' + y = 0 \quad \text{about } \underline{x=0}$$

$$P(x) = \frac{3x-1}{x(x-1)}$$

$$Q(x) = \frac{1}{x(x-1)}$$

$$p(x) = \frac{3x-1}{x(x-1)}$$

$$q(x) = \frac{1}{x(x-1)}$$

$$\lim_{x \rightarrow 0} x \cdot \frac{3x-1}{x(x-1)} = 1 = p_0$$

$$\lim_{x \rightarrow 0} x^2 \frac{1}{x(x-1)} = \lim_{x \rightarrow 0} \frac{x}{x-1} = 0 \quad \text{|| } q_0$$

• Indicial Eqn

$$r(r-1) + p_0 r + q_0 = 0$$

$$r^2 - r + r + 0 = 0, \quad r^2 = 0, \quad \underline{r = 0, 0}$$

* Let $y(x) = x^r \sum_{m=0}^{\infty} a_m x^m$ be solution of (1)

$$y'(x) = \sum_{m=0}^{\infty} (m+r) a_m x^{m+r-1} \quad y''(x) = \sum_{m=0}^{\infty} (m+r)(m+r-1) a_m x^{m+r-2}$$

$$\bullet \quad x(x-1)y'' + (3x-1)y' + y = 0$$

$$\Rightarrow \sum_{m=0}^{\infty} (m+r)(m+r-1) a_m x^{m+r} - \sum_{m=0}^{\infty} (m+r)(m+r-1) a_m x^{m+r-1} + 3 \sum_{m=0}^{\infty} (m+r) a_m x^{m+r} - \sum_{m=0}^{\infty} (m+r) a_m x^{m+r-1} + \sum_{m=0}^{\infty} a_m x^{m+r} = 0$$

$$\Rightarrow r(r-1) a_0 x^{r-1} + r a_0 x^{r-1} + \sum_{m=0}^{\infty} (m+r)(m+r-1) a_m x^{m+r} - \sum_{m=1}^{\infty} (m+r)(m+r-1) a_m x^{m+r-1} + 3 \sum_{m=0}^{\infty} (m+r) a_m x^{m+r} - \sum_{m=1}^{\infty} (m+r) a_m x^{m+r-1} + \sum_{m=0}^{\infty} a_m x^{m+r} = 0$$

$\downarrow$ $(m+1+r)(m+r) a_{m+1} x^{m+r}$ $\downarrow$ $(m+r+1) a_{m+1} x^{m+r}$

$$\Rightarrow (r^2 - r + r) a_0 x^{r-1} + \dots = 0$$

$$\underline{r^2 = 0}, \quad \underline{r = 0, 0}$$

Coefficient of x^{m+r}

$$+ 2(m+r) a_m$$

Coefficient of x^{m+k}

$$(m+k)(m+k-1)a_m - (m+k+1)(m+k)a_{m+1} + 3(m+k)a_m - (m+k+1)a_{m+1} + a_m = 0$$

$$\left[(m+k+1)(m+k) + (m+k+1) \right] a_{m+1} = \left((m+k)(m+k-1) + 3(m+k) + 1 \right) a_m$$

$$a_{m+1} = \frac{(m+k)^2 - (m+k) + 3(m+k) + 1}{(m+k+1)(m+k) + (m+k+1)} a_m$$

$$\rightarrow (m+k+1)(m+k+1)$$

$$\Rightarrow a_{m+1} = a_m \Rightarrow a_0 = a_1 = a_2 = a_3 = \dots$$

$$y_1(x) = x^0 \sum_{m=0}^{\infty} a_m x^m$$

$$y_1(x) = a_0 (1 + x + x^2 + x^3 + \dots) = \frac{a_0}{1-x}$$

$$y_2(x) = y_1(x) \ln(x) + x^2 \sum_{m=1}^{\infty} b_m x^m$$

$$y_2(x) = \frac{\ln(x) a_0}{1-x} + \sum_{m=1}^{\infty} b_m x^m$$